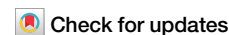


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Meta-analyses of the evolution of MXene synthesis for bioengineering and artificial intelligence-driven applications



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The year 2011 marked a breakthrough in material science innovation with the discovery of MXenes, an emerging family of two-dimensional transition metal-based nanomaterials. Owing to their distinctive properties, MXenes have rapidly surfaced as transformative materials, particularly in energy-storage and nanomedicine. In this review, we systematically explore the evolution of MXene synthesis, from its discovery to current advancements, focussing on their bioengineering applications, through a meta-analytic and bibliometric lens. We discuss synthesis methods, ranging from hydrofluoric acid (HF)-based etching to non-HF approaches, along with post-synthesis processes like intercalation, delamination and surface functionalization, that tailor MXene properties for biomedical therapeutics. We also overview key microscopy, spectroscopy and diffraction-based characterization methods, to understand their structure and functionality. Additionally, discussion on artificial intelligence (AI)-driven innovations highlights the significant shift in material science. By connecting synthesis methods with resulting characteristics and meta-analyses trends, this review emphasizes MXenes' transformative potential in regenerative therapeutics and diagnostics.

The development of materials science has progressed from the study of bulk materials to nanoscale systems, leading to the emergence of two-dimensional (2D) materials that enable the investigation of matter at the atomic scale. The exploration of 2D materials has been a transformative area of research within nanotechnology. The journey began with the discovery of carbon nanostructures, including carbon nanotubes (CNTs) and fullerenes, which ignited interest in isolating 2D materials from graphite¹. Graphene was experimentally isolated and characterized in 2004 as a single carbon-atom layer by Geim and Novoselov, which catalyzed a global interest in 2D materials^{2,3}. Consequently, graphene rapidly emerged as the prototype for a new class of 2D materials owing to its excellent electrical conductivity of approximately $1.5 \times 10^6 \text{ S m}^{-1}$ ⁴, an intrinsic tensile strength of about 130 GPa, and a Young's modulus close to 1 TPa⁵. Graphene sustains tensile strain up to approximately 6% under experimental conditions before mechanical failure^{6,7}. Researchers quickly extended their focus beyond graphene, exfoliating and exploring materials derived from layered precursors such as hexagonal boron nitride (h-BN), with in-plane thermal conductivities up to as high as $k_r \sim 550 \text{ W m}^{-1} \text{ K}^{-1}$ at room temperature⁸, dichalcogenides (e.g., MoS₂, WS₂, etc) with thermal conductivities of around $23.2 \text{ W m}^{-1} \text{ K}^{-1}$ ^{9,10}, and oxides¹¹. This expansion eventually included non-layered materials like

silicon, germanium, and tin, signaling the potential for a vast family of 2D compounds with diverse structures and properties.

A pivotal milestone in this journey was reached in 2011 with the discovery of MXenes at Drexel University, a novel class of 2D early transition metal-based carbides/nitrides/carbonitrides. MXenes were derived from the targeted etching of "A" layers in MAX phases, a group of layered ternary carbides and nitrides¹². MAX phases are denoted as $M_{(n+1)}AX_n$, where "M" denotes an early transition metal, "A" primarily denotes an element from the group 13 or 14, "X" is carbon and/or nitrogen, and "n" ranges from 1 to 4¹³. MAX phases are hexagonal, layered materials structurally classified in the P6₃/mmc space group. They consist of transition metal carbide/nitride octahedral sheets ("MX" layers) glued together with monoatomic layers of "A" elements^{14,15}.

The story of MAX phases began in the 1960s when Hans Nowotny and co-workers discovered approximately 50 such phases^{15–17}. In 1996, Michel Barsoum and Tamer El-Raghy synthesized single-phase Ti₃SiC₂, demonstrating its machinability and excellent electrical and thermal conductivity¹⁸. By 1999, the number of known MAX phases had expanded, and the term $M_{(n+1)}AX_n$ was introduced to encompass this family of materials¹⁹. By 2011, there were around 70 MAX phases, and this number increased to 155 in 2019²⁰. Today, more than 342 MAX phases have been discovered, offering a

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wide array of precursors for MXenes^{15,21–23}. The discovery of MXenes emerged from an unorthodox approach to reducing the dimensionality of MAX phases. Unlike other 2D materials, whose layered structures can be exfoliated due to weak van der Waals forces, MAX phases are unified by strong metallic or covalent bonds. This necessitated selective etching of the monoatomic A-layers while preserving the integrity of the MX layers. In a groundbreaking 2011 study, Yury Gogotsi and his team demonstrated that immersing Ti_3AlC_2 —a MAX phase with aluminum as the “A” element—in hydrofluoric acid (HF) selectively etched the aluminum layers at room temperature, leaving behind a 2D titanium carbide, Ti_3C_2 ¹². This process marked the birth of MXenes. Subsequent research showed that this method of focused “A” layer etching was relevant to other MAX phases. Additionally, the term “MXenes” was coined to reflect their connection to MAX phases and their reduced dimensionality. By 2012, researchers had established that MXenes could be synthesized from many other MAX phases, expanding the family significantly²⁴.

MXenes such as $\text{Ti}_3\text{C}_2\text{T}_x$ possess an exclusive amalgamation of properties, including high electrical conductivity (up to $24,000 \text{ S cm}^{-1}$)²⁵ and excellent thermal stability (above 1000°C under vacuum), while maintaining their structural integrity²⁶. Their intrinsic hydrophilicity, arising from surface terminations such as $-\text{O}$, $-\text{OH}$, and $-\text{F}$, facilitates easy dispersion in aqueous solutions and enhances interactions with biological systems²⁷. Furthermore, MXenes are chemically tunable, as their surface terminations can be selectively modified to tailor their physicochemical and interfacial properties^{28,29}, making them fit for diverse applications in energy storage, catalysis, water purification, and electronics³⁰. This discovery underscored the importance of interdisciplinary collaboration. As the field of 2D materials continues to advance, MXenes have become a cornerstone of research, bridging the divide between theoretical possibilities and practical applications. Future efforts aim to expand the library of MXenes, uncover new functionalities, and explore the synthesis-structure-property relationships that define their behavior.

In 2016, MXenes were first introduced into the biomedical field, marking a pivotal moment in their application. Lin et al. demonstrated their utility in photothermal cancer therapy, leveraging ultrathin $\text{Ti}_3\text{C}_2\text{T}_x$ nanosheets for tumor ablation under near-infrared (NIR) laser irradiation, supported by their exceptional photothermal conversion efficiency, biocompatibility, and low toxicity³¹. Rasoo et al. found that 2D $\text{Ti}_3\text{C}_2\text{T}_x$ exhibits antibacterial activity³². Our group demonstrated that MXenes can be

tailored to possess intrinsic immune-modulatory properties^{33–37}. Following this, MXenes have been increasingly explored across diverse biomedical domains, including drug delivery³⁸, bioimaging³⁹, immunomodulation³⁶, regenerative medicine³³, immunotherapy³⁶, bioelectronics⁴⁰, antibacterial coatings³², antimicrobial applications⁴¹, and tissue engineering⁴². These milestones reflect the remarkable versatility and potential of MXenes in addressing complex biomedical challenges, solidifying their role in advancing nanomedicine.

With this discussion on the comprehensive timeline of MXene synthesis since their discovery (Fig. 1), we will now explore different synthesis and post-synthesis processing, various characterization techniques, and highlight different types of transition metal-based MXenes (Fig. 2) in the upcoming sections of this review article. This review also examines how diverse synthesis methods contribute to the unique functional properties of MXenes, ultimately influencing their bioengineering applications. Furthermore, considering the significant advancements in MXene synthesis and applications, researchers continually strive to identify the optimal methods for producing robust MXenes with tailored properties for specific applications. In pursuit of this, we conducted two meta-analyses: the first examined the evolution in synthesis, characterization, and bioengineering applications of MXenes; the second focused on the integration of computational modeling and AI-driven predictive analytics approaches in MXene research. The findings from these analyses are mapped using VOS viewer and discussed in this review to deepen the understanding of current MXene synthesis trends, enabling the identification of emerging research areas and guiding future applications in bioengineering. Furthermore, the key transition metal-based MXene utilized in bioengineering applications is profiled individually, leveraging data from the first meta-analysis and complemented by bibliometric mapping. This comprehensive approach provides valuable insights into MXene research trajectories and the trends that have shaped the biomedical field over the past decade.

Synthesis of MXenes: etching techniques and post-synthesis processing

Synthesis of MXenes and the choice of an efficient synthesis method lay the foundation for the scope of MXenes to be employed in various applications. Since the discovery of MXenes in 2011, various researchers have ambitiously explored several methods to generate MXenes with unique characteristics and configurations (Fig. 2)⁴³. In order to develop the most efficient route of

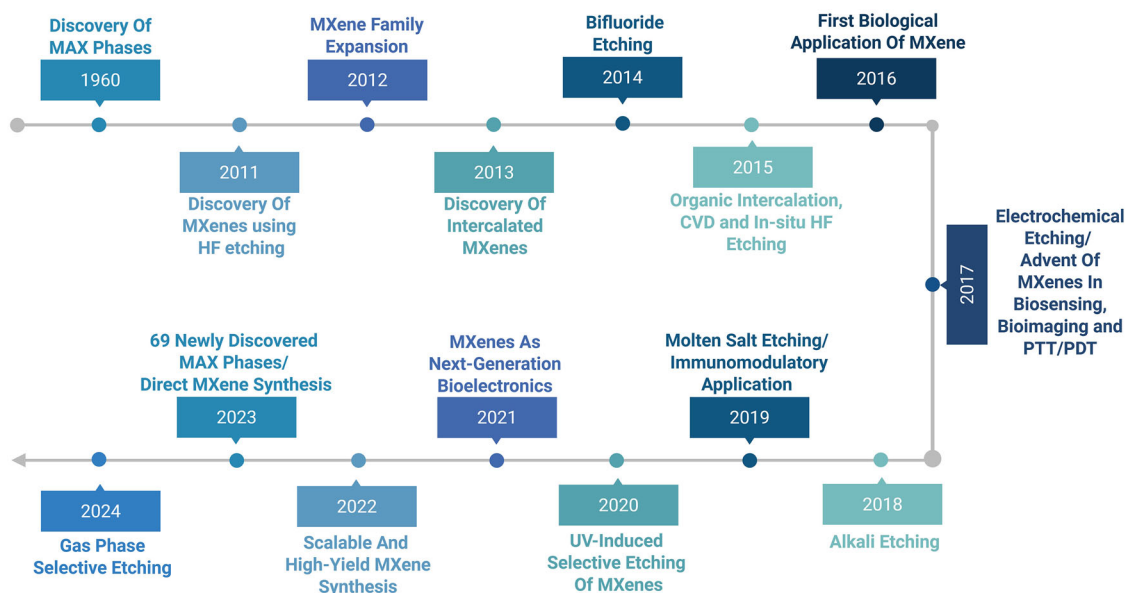


Fig. 1 | History and evolution of MXene synthesis and bioengineering applications (Created in BioRender). The timeline summarizes key milestones in MXene and MAX phase research, highlighting major developments across the years

1960^{16,17}, 2011¹², 2012²⁴, 2013⁸⁰, 2014²¹⁰, 2015⁴⁴, 2016³¹, 2017^{72,91}, 2018⁷⁴, 2019^{33,63}, 2020⁶⁸, 2021¹⁶⁶, 2022⁶⁶, 2023^{15,75}, and 2024^{76–78}.

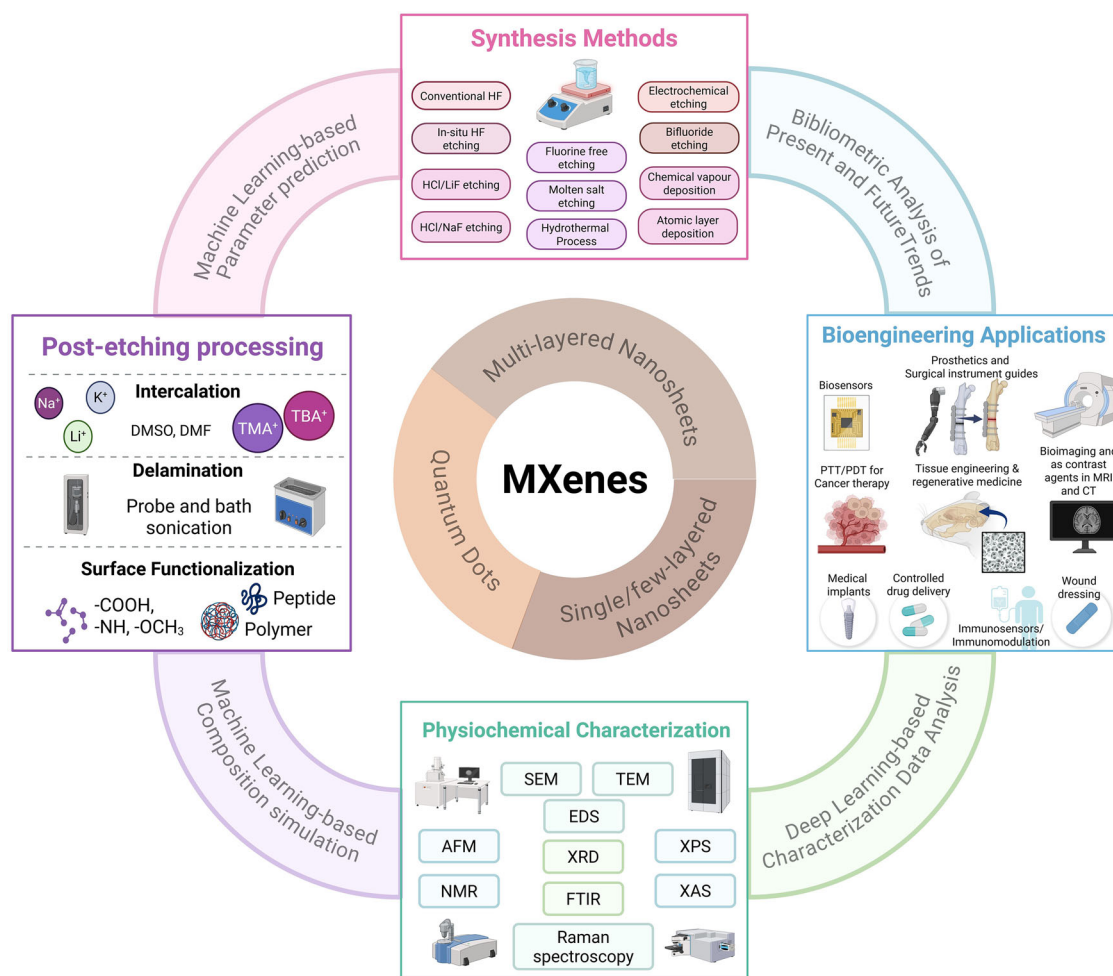


Fig. 2 | Integrative schematic overview of interconnected synergy between synthesis, post-etching processing, characterization, bioengineering application, and computation-based/predictive modeling of MXenes (Created in BioRender).

This figure provides a conceptual framework illustrating the interdependence of different aspects of MXene research from synthesis to biomedical applications, with computational approaches acting as integrative bridges across stages. The innermost concentric circle depicts the different forms of MXenes, including multi-layered nanosheets to single/few-layered nanosheets to quantum dots, representing the foundation for all subsequent processes and applications. Surrounding this, the outer circle represents the main elements covered in the review and their inter-connected flow in an anticlockwise direction: material synthesis to post-etching processing to physicochemical characterization to bioengineering applications. The top rectangular box summarizes various synthesis methods, including conventional HF-based and fluoride-free etching, bottom-up routes, physicochemical and

emerging methods such as electrochemical and molten-salt etching, and chemical vapor deposition. The left box outlines post-etching processing steps such as intercalation with ions or organic molecules, delamination via probe or bath sonication, and surface functionalization with chemical groups, peptides, or polymers to tailor properties for biomedical use. At the bottom, the physicochemical characterization box lists critical microscopy- and spectroscopy-based techniques essential for structural, compositional, and functional validation of MXenes. Encircling arcs emphasize how computational and analytical tools link these domains: machine learning-based parameter prediction connects synthesis with post-etching optimization; composition simulations bridge processing and characterization; and deep learning-based data analysis links characterization to bioengineering applications through structure–function correlations. Collectively, this framework highlights how computational integration deepens insights into MXene research, guiding optimization and emerging biomedical applications.

synthesis for a specific application with desired physicochemical properties, it is fundamental to fully apprehend the structure of the precursor, MAX phase¹⁵. The M-A bonds and the M-X bonds are conversely different, with weaker and stronger binding strengths, respectively⁴³. The former are held by metallic bonds, making it possible to selectively exfoliate the “A” layer using appropriate exfoliation methods called “etching”. The latter are held by covalent and/or ionic bonds. As a result of etching, multilayered-MXenes are generated. The earliest MXenes were synthesized by this selective exfoliation of “A” layers from the MAX, and hence are termed “selective etching”. Some of the less common types of MXene synthesis are categorized under “bottom-up synthesis,” like chemical vapor deposition (CVD)⁴⁴, solution-based synthesis⁴⁵, and atomic layer deposition (ALD)⁴⁶. From Fig.1, we see that over the years, different types of MXenes’ synthesis have evolved from different selective etching methods, including HF etching, in situ HF etching, electrochemical etching, and bottom-up synthesis for various

energy-based applications like synthesis of catalysis, electronics, sensors, optics, photodetectors, and bioengineering applications.

The HF-based etching is one of the earliest synthesis methods for producing MXenes, where HF selectively removes the “A” layer¹². Although this method is effective but it poses safety concerns due to the severely corrosive nature of HF⁴⁷, limiting its use in biological applications⁴⁸. Using HF as the etchant leads to the formation of defects, affects surface chemistry, influences material properties, and can result in Ti vacancies^{48–50}. HF removes some of the surface Ti atoms, thereby modifying the surface structure and impacting the conductivity of the resulting MXenes⁴⁸. Adjusting the HF concentration can mitigate defect levels and influence the surface functional groups. Higher HF concentration (from 5% up to 30%) increases the proportion of -F terminations and provides more H⁺ ions that react with oxygen-containing functional groups like -OH and -O, leading to the formation of H₂O. This process creates additional surface vacancies that

are subsequently filled by -F, resulting in an increase in the number of -F surface terminations^{48,51,52}. MXenes with abundant -O terminations and -OH terminations exhibit enhanced capacitance⁵³, stability⁵⁴, and electrochemical behavior⁵⁵. In contrast, excess-F terminations, which are a consequence of the HF etchant route, limit these properties. Therefore, lower HF concentrations are preferable for producing MXenes via this route to decrease the presence of defects and undesirable surface terminations. The specific concentration of HF required also depends on the type of transition metal in the MAX phase. Titanium-based MXenes can be produced using low HF concentrations, as discussed. In contrast, vanadium and niobium-based MXenes require higher HF concentrations (up to 50%), which results in the predominance of -F surface terminations^{52,56}. To mitigate the hazardous outcomes of HF, researchers have developed alternative etching strategies using in situ HF generation to remove “A” layer. Here, by mixing acids like hydrochloric acid (HCl) with fluoride salts, including potassium fluoride (KF), sodium fluoride (NaF), and lithium fluoride (LiF), this method improved the biosafety of the synthesized MXenes and facilitated improved control over surface terminations⁵⁷. In some cases, a combination of strong acids like HCl and H₂SO₄ with HF can be used for MXene synthesis⁵⁸. MXenes etched using HF/HCl have higher amounts of -O and -Cl surface terminations and exhibit larger interlayer spacing due to the presence of more structural water. This increase in structural water enhances the electronegativity of the resulting MXenes. In contrast, MXenes etched using HF/H₂SO₄ exhibit higher conductivity, thermal stability, and overall electrochemical performance attributed to the effective elimination of etching derivatives and structural water during this process^{58,59}. Elsewhere, oxidative etching using an aqueous system of H₂SO₄ and hydrogen peroxide (H₂O₂) as the etchant was employed to obtain highly -O and -OH terminated MXene nanosheets. This method is reported to be a safe and non-toxic etching route with no harmful by-products⁶⁰. Meanwhile, HF-based bifluoride salts⁶¹ such as ammonium hydrogen fluoride (NH₄HF₂)⁴⁶, or a mixture of fluoride-based salts⁶¹, can also be used to etch the “A” layer. However, the only drawback of these methods is the introduction of fluoride-based surface terminations that may pose safety concerns in biological systems and negatively affect the chemical stability and reactivity of MXenes.

Recent advancements in MXene synthesis include the use of fluoride-free etching methods to address any environmental and biological safety concerns that the HF/ in situ HF methods pose. Using molten salt etching methods has emerged as a promising alternative to remove “A” layer without introducing fluoride-based surface terminations, with increased applicability, scalability, and control over surface functional groups⁶². The first fluorine-free, water-free, green synthesis of MXenes was reported in 2019 using Lewis acid zinc chloride (ZnCl₂) molten salt^{63–65}. Since then, several series of MXenes from Titanium (Ti), Niobium (Nb), Tantalum (Ta) based MXenes have been able to be synthesized directly from the MAX phase via the molten salt etching method⁶⁶. Although this method provides a broad etching range and ensures chemical and biosafety, it is still in the early stage of development. Further investigations are needed to fully explore and optimize the accordion-like structure, as well as the physical and chemical characteristics of the resulting MXenes⁴³. While theoretical models predict outcomes based on the starting elements and surface terminations, experimental results often show more complex surface group combinations, creating a significant gap between theoretically predicted characteristics and the actual behavior observed. Other physiochemical methods of synthesis include using halogens⁶⁷, ultraviolet light⁶⁸, mechanical/electromagnetic waves⁶⁹, and thermal reduction⁷⁰ strategies have been reported for MXene synthesis. By harnessing different energy sources, these techniques offer alternatives to conventional etching methods to facilitate MXene formation. Beyond these methods, some of the emerging MXene synthesis methods include electrochemical etching, alkali-based etching, direct synthesis, and gas-phase synthesis⁷¹. Electrochemical etching was first reported using a diluted hydrochloric acid (HCl) aqueous electrolyte, where Al was selectively removed from porous Ti₂AlC electrodes, resulting in the formation of a Ti₂CT_x MXene layer directly on the Ti₂AlC substrate⁷². In another study,

aqueous tetrafluoroboric acid (HBF₄) was employed as the electrolyte to electrochemically synthesize Ti₃C₂ and Ti₃CN MXenes⁷³. This method substantially reduces HF usage by relying on the in situ generation of a low concentration of HF during the etching, thus offering a safer and more controlled alternative to traditional HF-based methods. Alkali-assisted synthesis serves as a fluoride-free alternative to traditional HF-based MXene production⁷⁴. In this approach, strong alkaline solutions such as NaOH are used to selectively etch the Al layers from the MAX phase precursor. Initial challenges involving the production of insoluble byproducts, including aluminum hydroxide, reducing the yield of MXenes were addressed by subsequent process improvements, inspired by bauxite refining techniques. This enabled better dissolution of these byproducts, leading to the successful extraction of multilayered Ti₃C₂T_x MXenes with higher purity. Though this method provides a greener and safer synthesis route compared to fluoride-based techniques, further optimization is needed to improve efficiency, scalability, and material quality. Direct synthesis methods, such as molten-salt electrochemical etching, have enabled the generation of Ti₂C MXene directly from elemental precursors⁷⁵. Additionally, solvent-free, fluorine-free, gas-phase approaches offer scalable and safer routes for the mass production of -Cl group terminated MXenes (Ti₃C₂Cl₂)^{76–78}. Although some of these approaches are in early stages, they represent important progress in the development of MXene synthesis strategies. Another important aspect of the synthesis of MXenes that significantly determines their properties for tailored applications is the post-synthesis processing, including intercalation, delamination, and surface functionalization⁴⁵. The use of strong acids with fluoride salt or bifluoride salts, or fluoride-based salt for MXenes synthesis facilitates the introduction of intercalating metal ions (Li⁺, Na⁺, K⁺) / organic molecules between the MXene layers⁷⁴. Inorganic intercalation through the insertion of metal ions weakens the interlayer forces, allowing the multilayered MXenes to be transformed into single-layer MXenes during the subsequent delamination process⁷⁹. This results in enhanced electrical conductivity, as well as ion/catalytic properties, making MXenes ideal for energy storage, biosensors, and biocatalysis. Organic intercalation using N,N-dimethylformamide (DMF), dimethyl sulfoxide (DMSO), hydrazine monohydrate (HM), urea, tetrabutylammonium hydroxide (TBAOH), and tetramethylammonium hydroxide (TMAOH) is utilized depending on the intended application of the synthesized MXenes^{79–81}. After intercalation with these organic solvents, single-layer MXenes colloids can be collected from the supernatant through a centrifugation process. Furthermore, delamination is a process of separating the multilayered MXenes into single or few-layered nanosheets, typically achieved through mechanical techniques like sonication^{80,82}. Mechanical delamination separates the layers of multilayered MXenes through longitudinal/transverse stress⁴³. This process increases surface exposure and reduces interlayer interactions, making MXenes well-suited for applications in sensing, drug delivery, and photocatalysis. Additionally, a cyclic freeze-thaw approach is employed, where the freezing process expands the interlayer distance, due to the expansion force from water freezing, resulting in significant exfoliation of MXenes with increased lattice spacing⁸³. Beyond multilayered to few/single-layered nanosheet, MXene quantum dots (MQDs) represent the zero-dimensional derivatives that exhibit distinct optical and electronic properties arising from quantum confinement effects⁸⁴. To obtain MQDs from MXenes, top-down routes predominate, including hydro^{33,36,41} thermal/solvothermal^{85,86} cleavage, ultrasonic fragmentation⁸⁷ (bath or probe), electrochemical routes^{88,89}, ball-milling⁹⁰, and hybrid sequences^{87,91} (e.g., intercalation-plus-sonication or ultrasonic-hydrothermal). All these methods disrupt 2D sheets into quantum dots while simultaneously inducing surfaces with functional terminations⁹². Specialized physical methods have also emerged, such as acoustomicrofluidic processing at room temperature, yielding pristine MQDs^{93,94}, and laser-based strategies (including femtosecond shaped-laser and liquid-phase laser ablation) that enable HF-free production directly from MAX precursors at useful yields^{93,95}. In contrast, bottom-up synthesis strategies, though less explored, construct MQDs from molecular or ionic precursors rather than breaking down sheets. These include molten-salt synthesis⁹⁶ and

pyrolysis⁹⁷, which offer precise control over composition, crystallinity, and size, but remain challenging for large-scale production.

The “T_x” surface terminations of MXenes contribute to their characteristic physicochemical properties. These surface-terminating functional groups like hydroxyl (-OH), fluorine (-F), and oxygen (-O) provide active sites that impart hydrophilicity, drug loading capacity, and enable further surface modification to improve bioavailability⁹⁸. As a result, surface functionalization has garnered significant attention for improving its suitability in various biological applications. These surface terminations enable the integration of specific functional groups, such as amines, peptides, or polymers, to functionalize MXenes⁹⁹. This improves their stability, dispersibility, and biocompatibility, while also enhancing interactions with biomolecules, thereby promoting applications like drug delivery, biosensing, and tissue engineering¹⁰⁰. The hydrophilic nature of functionalized MXenes facilitates better cell interaction and controlled drug release, while their high surface area, tunability, and conductivity make them ideal for bioelectronics⁴⁰. In addition, functionalized MXenes can improve targeting efficiency and reduce cytotoxicity, addressing challenges in biomedical applications by extending their antioxidant, antibacterial, and anti-inflammatory properties¹⁰¹.

Further, to systematically evaluate the evolution of MXene synthesis and its applications in bioengineering fields, we conducted a structured meta-analysis using the Scopus database. Our search strategy (Fig. 3A) employed the keywords “MXene” AND (“synthesis” OR “etching” OR “functionalization”) AND (“biomedical” OR “bioengineering” OR “tissue engineering” OR “photothermal” OR “immunomodulatory”), with Boolean logic applied to capture studies linking synthesis approaches to therapeutic and bioengineering applications. The search was filtered to include publications from 2015 to 2025, limited to articles and reviews published in English. This search yielded 225 articles and 222 reviews, with a notable surge in research over the last four years (Fig. 3B). The findings of this analysis revealed a dataset containing 478 items grouped into 6 clusters, forming 37,510 links with a total link strength of 86,847. This dataset was analyzed to guide the subsequent in-depth discussion on characterization techniques, material properties, and bioengineering applications of different transition metal-based MXenes. Bibliometric evaluation was further performed using VOS Viewer, selecting “co-occurrence” as the analysis type and “author keywords” as the unit of analysis. Keyword occurrences were assessed using the full counting method, ensuring equal weighting for each keyword. To normalize relationships and enhance interpretability, the association strength method was applied. Density visualization using VOS Viewer (Fig. 3C) was then used to represent these findings and highlight the current research trends, and provided a foundation for driving future biomedical innovations. In addition, building on the meta-analysis results, bibliometric network mapping was conducted using VOS Viewer to examine the interconnections among the different types of transition metal-based MXenes employed in the synthesis (Fig. 4A), their physicochemical characterization (Fig. 4B) and the correlations with specific bioengineering and therapeutic applications (Fig. 4C). Titles, abstracts, and keywords from the curated dataset were imported into VOS Viewer to construct co-occurrence and network visualizations, where nodes represented key concepts or applications and links reflected the strength of their co-occurrence. This approach revealed thematic relationships, conceptual clusters, and emerging trends across synthesis, characterization, and biomedical application domains. Together, the overview of the synthesis and post-synthesis processing of MXenes discussed, along with the meta-analysis results, sets the stage for the upcoming sections, which will explore how MXenes can be characterized and how each synthesis method influences their biological properties for use in biomedical applications.

Advanced MXene characterization: Bridging structure and physicochemical properties

A comprehensive characterization of MXenes involves integrating multiple analytical techniques to provide complementary evidence about their morphology, structure, and surface chemistry, paving the way for their

diverse applications^{102,103}. The different types of material characterization methods include scanning electron microscopy (SEM), energy dispersive X-ray spectroscopy (EDS), transmission electron microscopy (TEM), X-ray photoelectron spectroscopy (XPS), X-ray absorption spectroscopy (XAS), electron energy loss spectroscopy (EELS), Fourier-transform infrared spectroscopy (FTIR), X-ray diffraction (XRD), atomic force microscopy (AFM), Raman spectroscopy and nuclear magnetic resonance (NMR).

Recently, Secondary Ion Mass Spectrometry (SIMS) has emerged as a valuable addition to the characterization technique, enabling atomic-level analysis of both MAX phases and MXene.

The SEM and TEM enable the examination of the structure, arrangement, and layers of the MAX phases and MXenes. The appearance of “accordion-like” structure seen in SEM images helps differentiate between the MAX phases and MXenes. The EDS is used in complement to SEM, providing a quantitative report about the elemental composition of the MAX phases and MXene. Specifically, understanding the elemental percentage of aluminum between the MAX phase and MXenes offers additional insight about the degree of successful etching and the extent of accordion-like structures observed in the SEM images (Fig. 5). For instance, in the study of Ti₃C₂T_x MXene nanosheets, by Yan et al.³⁷, SEM provided insights into the two-dimensional morphology, including layer separation and interlayer spacing, while EDS mapped the elemental composition, confirming the occurrence of titanium, carbon, oxygen, and fluorine as the main elements in Ti₃C₂T_x MXene (Fig. 5h, j). These findings were further supported by TEM, which showed an explicit planar geometry and further confirmed the hexagonal crystalline structure through fast Fourier transform (FFT) (Fig. 5l) and selected area electron diffraction (SAED) (Fig. 5m) analyses.

The surface terminations, electronic structure, and surface chemistry of MXenes are characterized by analytical techniques like XPS, FTIR, XAS, and EELS. XPS is effective for identifying surface terminations (-OH, -O, -F) and chemical states of the elemental atoms, which influence properties such as hydrophilicity, solubility, and biocompatibility, as seen in amino-functionalized MQDs¹⁰⁴ for biomedical applications¹⁰³. The FTIR¹⁰⁴, XAS¹⁰⁵, and EELS¹⁰⁶ complement the surface-sensitive XPS by providing detailed information on surface chemistry, valence states, and electronic structure. Combining these techniques addresses the limitations of individual methods and enables comprehensive characterization of MXenes’ surface chemistry (Fig. 6A).

The X-ray diffraction (XRD) plays a pivotal role in characterizing the crystal structure, confirming the effective etching of Al and phase identity of MXenes and their derivatives, such as MXene quantum dots (MQDs)^{36,104}. MXene sheets typically exhibit a crystalline structure, with the metal atoms arranged in a hexagonal close-packed (HCP) lattice and non-metal atoms residing in the octahedral positions between the HCP layers. In a study by Yan et al., XRD analysis of Ti₃C₂T_x MXenes reveals a characteristic peak near 8° 2θ, corresponding to the (002) plane, indicative of a layered crystalline structure with negligible aluminum layers³⁷. XRD, when combined with TEM, provides macroscopic and microscopic insights into the material’s structure, yielding more information than either technique alone.

The AFM is extensively used for characterizing two-dimensional (2D) materials, including MXenes, due to its ability to provide three-dimensional (3D) information about the lateral dimensions and vertical thickness of flakes at sub-nanometer resolution¹⁰⁷. AFM offers certain advantages over electron microscopy and optical methods, such as the ability to measure both lateral dimensions and thickness accurately. AFM is frequently combined with Raman spectroscopy to offer a comprehensive characterization of MXenes¹⁰⁸. While AFM provides precise measurements of lateral dimensions and vertical thickness, Raman spectroscopy complements this by offering insight into the material’s vibrational modes and surface chemistry. This combined approach enhances the understanding of the MXenes’ structural and chemical properties, enabling accurate assessments of both morphology and functionality. Figure 6B displays Raman spectroscopy analysis of the structure, morphology, and surface chemistry of Ti₃C₂T_x MXenes of different forms, synthesized using different etchants and

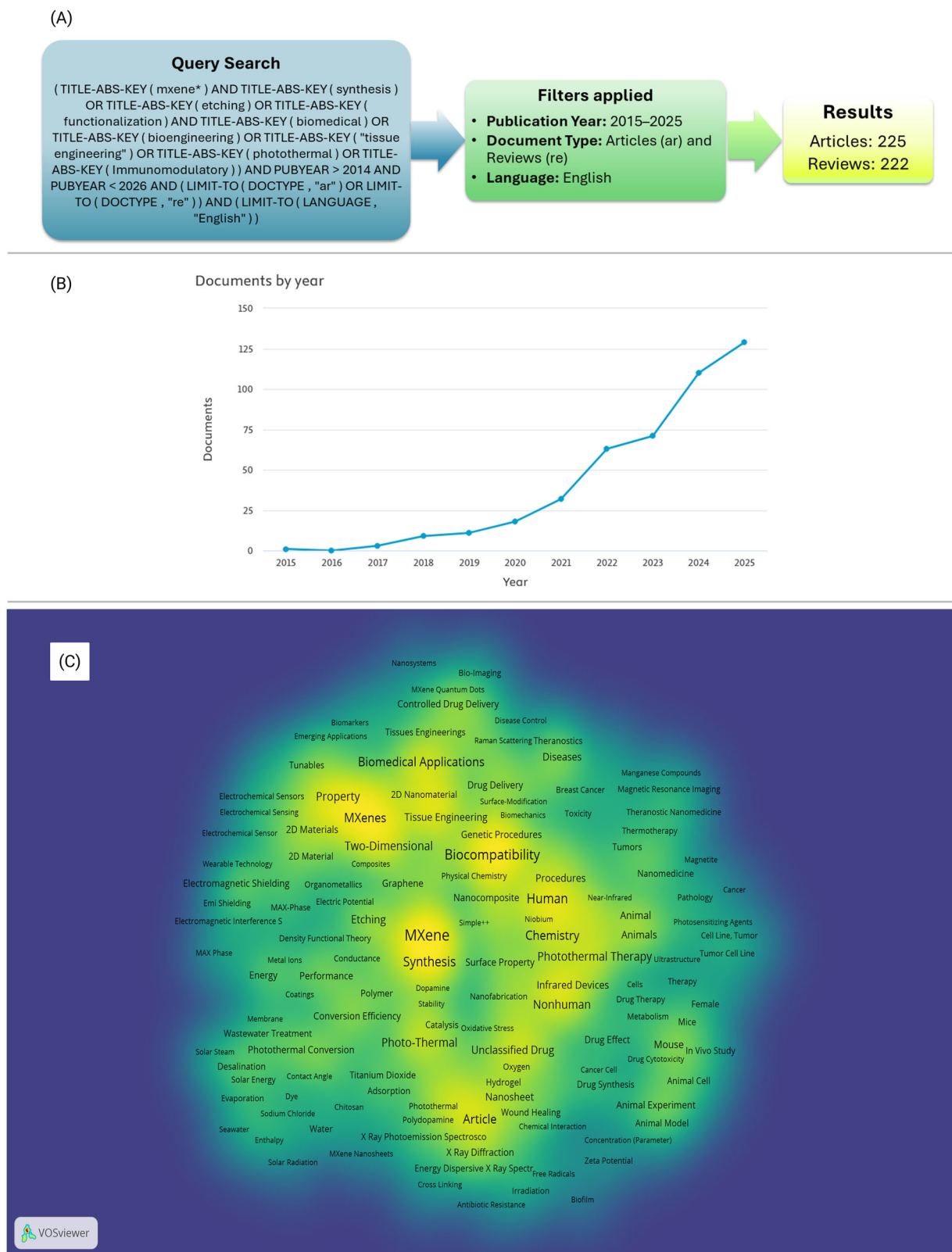
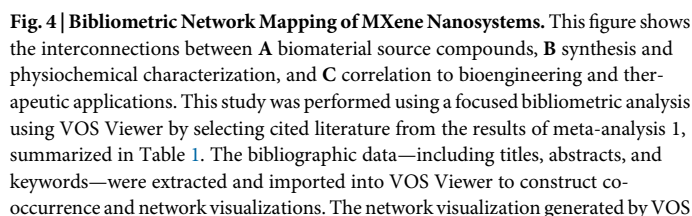


Fig. 3 | Overview of the meta-analysis conducted on the evolution of MXene synthesis and applications, with emphasis on their application in bioengineering fields. A Flowchart illustrating the literature search strategy, starting from the initial query and progressing through the applied filters. The query search was limited to articles and reviews published in English between 2015 and 2025, yielding 225 articles and 222 reviews; **B** Distribution of publications by year over the last decade;

C Density visualization of keywords co-occurrence and thematic clusters in literature relating to synthesis, material characterization, and applications. The bibliometric evaluation involved VOS Viewer software, with the analysis type set to “co-occurrence” and “author keywords” chosen as the unit of analysis, allowing insight into the structure and research trends within the field.



Viewer displays nodes representing key concepts, compounds, or application areas, while the links between nodes indicate the frequency and strength of their co-occurrence across the curated literature set. The link strength quantifies how often two terms or items appear together, with thicker or more numerous links denoting stronger thematic relationships. This approach provided an overview of the interconnectedness of research themes such as synthesis methods, characterization techniques, and application domains in biomaterials, while enabling mapping of conceptual clusters and emerging trends.

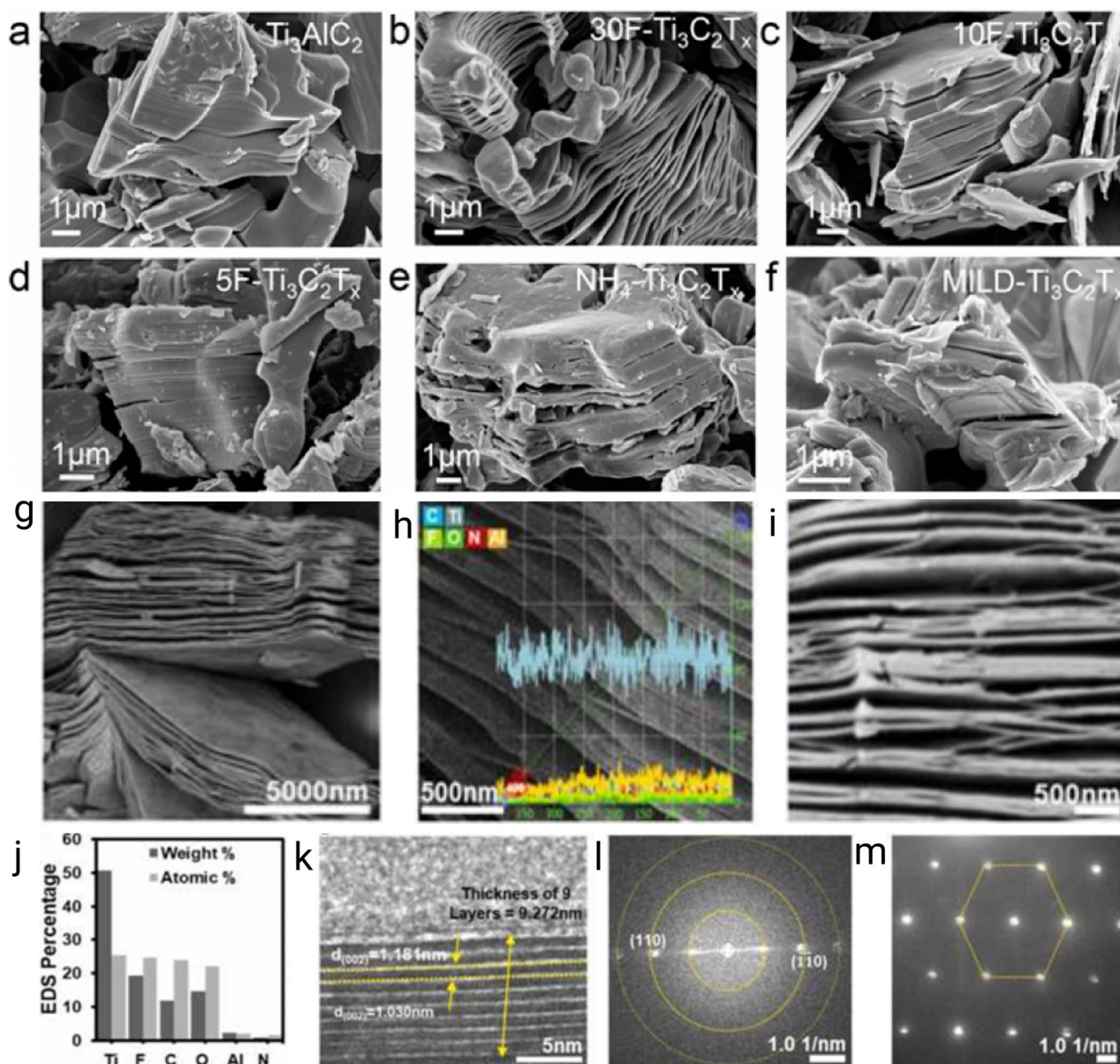


Fig. 5 | Microstructural characterization of $\text{Ti}_3\text{C}_2\text{T}_x$ MXene nanosheets using SEM and TEM analysis. SEM image of **a** Ti_3AlC_2 (MAX) powder showing the compact layered structure. Multi-layered $\text{Ti}_3\text{C}_2\text{T}_x$ MXenes etched using **b** 30 wt % HF, **c** 10 wt % HF, and **d** 5 wt % HF **e** NH_4HF_2 and **f** LiF/HCl , showing the extent of accordion-like morphology. (Reprinted (adapted) with permission from ref. 57 © 2017, American Chemical Society). **g** MXene nanosheets with 150 nm interlayer spacing. **h** EDS line scan of MXene nanosheets reveals Ti, C, O, and F as the

dominant elements. **i** Backscattered electron (BSE)-SEM image of $\text{Ti}_3\text{C}_2\text{T}_x$ MXenes. **j** EDS area scan of the $\text{Ti}_3\text{C}_2\text{T}_x$ MXene nanosheets, presenting the atomic and weight percentage of different elements in the elemental makeup of MXene flakes. **k** HRTEM image of MXenes showing a thickness of 9 layers is 9.272 nm, **l** FFT and **m** SAED analysis of $\text{Ti}_3\text{C}_2\text{T}_x$ MXene nanosheets verified the presence of a hexagonal crystalline structure (Reproduced with permission from ref. 37 © 2022, The Author(s). Published by Elsevier Ltd).

the effect of different delamination methods¹⁰⁸. Raman spectroscopy, a vibrational spectroscopic technique, measures the inelastic scattering of light, providing detailed information for studying MXenes' molecular fingerprints. Furthermore, Raman spectroscopy is useful for characterizing MXenes in composite materials, offering insights into their dispersion, stability, and interaction with other components, which is essential for designing advanced materials for energy storage, sensing, and other high-performance applications¹⁰⁹. The NMR also plays a crucial role in analyzing low-mass elements, such as hydrogen and fluorine, on the MXene surface. NMR helps determine the ratio of surface terminations, including -OH, -O, and -F, which are influenced by the synthesis method¹⁰². Additionally, NMR can reveal water molecules bound to the MXene surface and track ion interactions and swelling in different environments^{110,111}. The shifts in

chemical signals, such as the ^{13}C chemical shift between $\text{Ti}_3\text{C}_2\text{T}_x$ and V_2CT_x , are also detected, providing insight into the changes in electronic properties of MXenes¹¹².

Secondary Ion Mass Spectrometry (SIMS) is a highly sensitive characterization method in which sputtered secondary ions are separated and detected according to their mass-to-charge ratios^{113,114}. It is used for the compositional analysis of MXenes and their precursor MAX phases, providing depth profiles at the atomic-layer scale, which enables quantitative determination of elemental composition and distribution across layers. In a study by Michałowski et al. (2022), ultralow-energy SIMS was used to identify oxygen incorporation into carbon sublattices, confirming the existence of oxycarbide MAX compositions and determining the composition of adjacent termination structures in resultant MXenes¹¹³. This

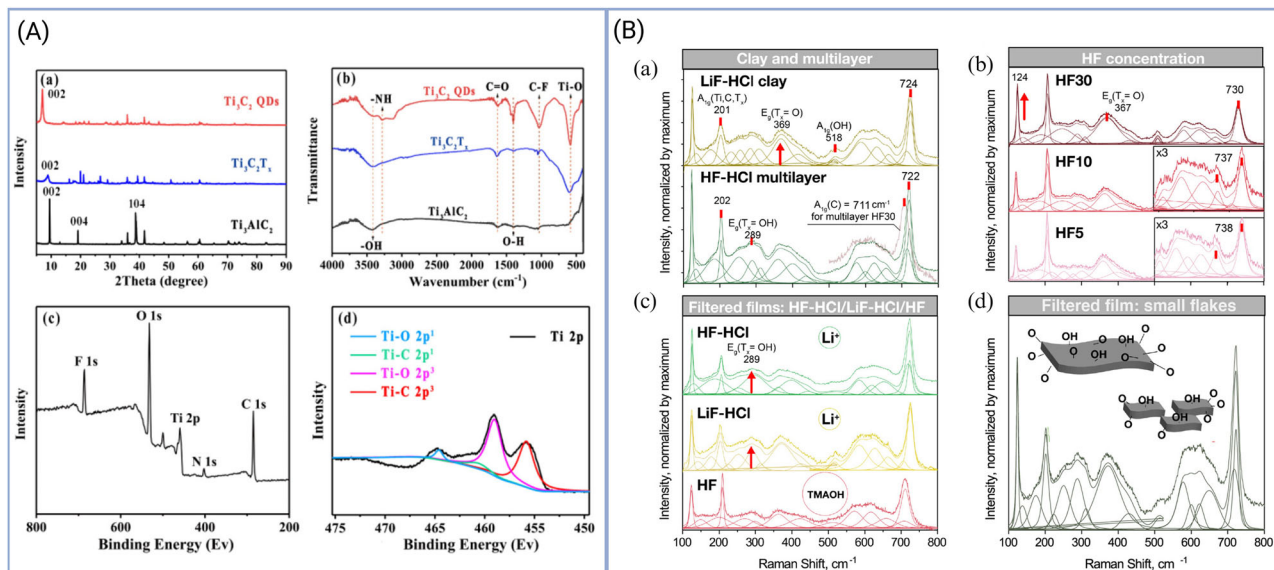


Fig. 6 | Spectrometry-based characterization of MAX phases and MXenes.

A Surface chemistry and crystallinity analysis of MAX phase, MXenes, and MQDs. (a) XRD spectra of Ti_3AlC_2 MAX Phase, multi-layered $\text{Ti}_3\text{C}_2\text{T}_x$ MXene, and $\text{Ti}_3\text{C}_2\text{T}_x$ MQDs. (b) FT-IR spectra of Ti_3AlC_2 , $\text{Ti}_3\text{C}_2\text{T}_x$ MXene, and $\text{Ti}_3\text{C}_2\text{T}_x$ MQDs. (c) Full Scan and (d) High-resolution XPS spectra of Ti 2p for Ti_3C_2 MQDs (Reprinted (adapted) with permission from ref. 104 © 2021, American Chemical Society). **B** Influence of etching solutions and intercalants/delamination methods on vibrational modes of $\text{Ti}_3\text{C}_2\text{T}_x$ MXene analyzed using Raman Spectroscopy. Deconvolution was carried out using the Voigt function in WiRE 3.0 software, and for every

deconvoluted spectrum shown, a matrix of 100 spectra was collected and consolidated statistically. Raman spectra of (a) clay and multilayer $\text{Ti}_3\text{C}_2\text{T}_x$ MXenes. (b) $\text{Ti}_3\text{C}_2\text{T}_x$ MXenes etched with different concentrations of HF (c) $\text{Ti}_3\text{C}_2\text{T}_x$ MXenes synthesis using different etchants and intercalation methods, where for HF-HCl and LiF/HCl etchants, Li ions were intercalated, and for the HF etchant, TMAOH was used as an intercalant. (d) Raman spectrum of $\text{Ti}_3\text{C}_2\text{T}_x$ flakes with size = 100 nm, etched with HF-HCl and filtered into a film (Reprinted (adapted) with permission from ref. 108, © 2020, American Chemical Society).

investigation revealed that many MXenes possess significant oxygen content, categorizing them as oxycarbide MXenes rather than pure carbide analogs. Based on the SIMS results, the authors suggest that many previously reported MXene properties may require re-evaluation, as recognizing that a substantial number of conventionally synthesized MXenes are in fact oxycarbides provides a clearer explanation of inconsistencies in the literature. While oxygen-enriched surfaces can enhance the physicochemical properties like hydrophilicity, biocompatibility, and conductivity^{115,116}, excessive oxygen incorporation can induce lattice distortions that potentially compromise the mechanical stability and electrical performance of MXenes^{117–119}. Although the role of oxygen substitution in altering MXene characteristics is being studied, its effect on the properties of the associated precursor MAX phases remains less explored¹²⁰. Research by Anayee and colleagues addresses this shortcoming by highlighting that the successful production of MXenes critically relies on the quality and characteristics of precursors used for the MAX formation¹²¹. This study investigates how synthesis variables and oxygen incorporation in the lattice affect the oxidation resistance of Ti_3AlC_2 MAX powders. Substitutional oxygen has been shown to degrade the resulting MXenes through oxidation and hydrolysis, diminishing their electrical conductivity. Combined XRD and thermogravimetric analysis (TGA) confirm that removing substitutional oxygen from the MAX lattice can reduce oxidation resistance. Although MAX phases are typically derived from non-oxide precursors, oxygen contamination occurs during the synthesis, exhibiting carbon deficiencies and oxygen substitutions forming oxycarbides with the general formula MC_xO_y ^{122,123}. A parallel study refined SIMS data deconvolution and calibration protocols, enabling layer-by-layer compositional analysis of MAX phase and MXene samples with ± 1 % accuracy¹²⁴. The analysis revealed that the X layer of carbide, nitride, and carbonitride MXenes commonly contains substantial oxygen, leading to the revolutionary recognition and confirmation of oxycarbide, oxynitride, and oxycarbonitride subfamilies, respectively. These insights underscore the critical importance of achieving high phase purity in the precursor MAX phases, as impurities or mixed-phase precursors complicate MXene synthesis and performance. Therefore, precise MAX phase synthesis^{15,21}

methods like hot pressing, spark plasma sintering, and the use of thorough characterization techniques like XRD¹²⁰, TGA¹²¹, and SIMS¹¹³ are essential for producing phase-pure MAX phases to yield consistent MXene morphology and functional properties.

Furthermore, tip-enhanced Raman spectroscopy (TERS), AFM, SEM, and TEM offer nanoscale resolution of the number of layers in MXenes, defined by the structural parameter “ n ” in the formula $\text{M}_{n+1}\text{X}_n\text{T}_x$, which critically determines their physicochemical, electronic, mechanical, and biological properties^{125,126}. Precise characterization of layer thickness is vital for understanding how MXenes can be tailored for specific bioengineering applications, as this parameter strongly impacts both their physical and biological properties^{127,128}. Lower- n MXenes (e.g., M_2XT_x , such as Ti_2CT_x ^{129–131} and V_2CT_x ^{132–134}) possess higher surface-to-volume ratios and increased density of surface terminations, which enhance dispersibility^{132,135} compared with the widely studied $\text{M}_3\text{X}_2\text{T}_x$ systems such as $\text{Ti}_3\text{C}_2\text{T}_x$. These attributes favor rapid charge transfer, high surface reactivity, and biointeraction, making lower- n MXenes favorable for applications like drug delivery, photothermal therapy, and biosensing. Higher- n MXenes (e.g., $\text{M}_4\text{X}_3\text{T}_x$, $\text{M}_5\text{X}_4\text{T}_x$, such as $\text{Ta}_4\text{C}_3\text{T}_x$ ³⁶, $\text{Ti}_4\text{N}_3\text{T}_x$ ^{61,136,137}, and $\text{V}_4\text{C}_3\text{T}_x$ ^{138,139}), on the other hand, offer improved electrical conductivity, mechanical robustness, and stability, making them better suited for immunomodulation, tissue scaffolding, coatings, and implantable devices⁴⁰. Comparative studies of Ti_2CT_x and $\text{Ti}_3\text{C}_2\text{T}_x$ MXenes have shown that increasing “ n ” results in thicker, stiffer nanosheets with altered elastic and magnetic properties¹⁴⁰, and modulated antibacterial activity⁴¹, highlighting the importance of layer number in determining the structural and biofunctional behavior of MXenes¹⁴². Likewise, titanium nitride MXenes reveal that increasing “ n ” can decrease in-plane Young’s modulus while enhancing stability, underscoring the interplay between surface reactivity and mechanical performance¹⁴³. This study addressed two complementary aspects of comparative analysis of carbide versus nitride-based MXenes and the influence of “ n ” within the nitride MXenes. Nitride-based MXenes possessed reduced lattice constants and monolayer thickness than carbide-based MXenes. Their in-plane Young’s modulus is larger than carbide-based MXenes, yet, in both families,

this modulus decreases as the number of layers increases. was inversely proportional to monolayer thickness. Cohesive energy calculations indicated that larger monolayer thickness enhances the structural stability of MXenes, while adsorption energy calculations revealed that nitride-based MXenes exhibit stronger affinity for surface terminations, yielding more chemically active surfaces compared to carbide counterparts. Notably, nitride MXenes also exhibited higher electrical conductivity than carbides, showing their growing potential for applications that require both high conductivity and surface chemistry.

Complementing *ex situ* imaging, *in situ* approaches enable real-time monitoring of structural, morphological, and chemical changes in MXenes under synthesis or working conditions. For mechanical properties, *in situ* tensile testing within a TEM, combined with AFM nanomechanical mapping, revealed that $\text{Ti}_3\text{C}_2\text{T}_x$ nanosheets exhibit a Young's modulus of approximately 80–100 GPa perpendicular to the basal plane and can sustain tensile strength up to 670 MPa for 40 nm thick nanoflakes¹⁴⁴. *In situ* electron energy loss spectroscopy (EELS) combined with extended energy loss fine structure (EXELFS) analysis, synchrotron X-ray absorption fine structure (XAFS), and density functional theory (DFT) calculations were employed to analyze the surface functionalization and thermal stability of $\text{Cr}_2\text{TiC}_2\text{T}_x$ ¹⁴⁵. This integrated approach revealed that vacuum annealing up to 600 °C results in the complete removal of –F terminations, thereby providing a pathway for controlled termination engineering to tune the magnetic and electronic properties of MXenes. *In situ* X-ray absorption spectroscopy (XANES) and extended X-ray absorption fine structure (EXAFS) monitored electronic and structural changes during hydrogen evolution reaction (HER) in $\text{Ti}_4\text{N}_3\text{T}_x$ and alloyed $\text{MTi}_4\text{N}_3\text{T}_x$ ($M = \text{Mn, Mo, Cr}$) MXenes¹⁴⁶. XANES showed multiple Ti oxidation states, with Cr partially reduced from Cr^{3+} to Cr^{2+} under HER, while Mn and Mo remained unchanged. EXAFS revealed coordination changes, including modifications in Mo–O, Mo–Ti, and Cr–O bonds, indicating oxygen vacancy formation under reaction conditions. These results demonstrate that HER activity correlates with dynamic changes in metal oxidation states and coordination environments, where oxygen vacancies on basal planes serve as catalytically active sites. These advanced *in situ* techniques enhance our understanding of MXenes by capturing structural, mechanical, and chemical changes as they occur.

Thus, the integration of physicochemical characterization methods and complementary techniques, as discussed above, facilitates a thorough understanding of the structural, chemical, and electronic properties of MXenes, permitting their optimization for various applications.

AI-driven predictive modeling and bibliometric analysis of MXene Synthesis and functional applications

Given the rapid advancements in artificial intelligence (AI) technologies across all scientific fields, their application in MXene research has significantly expanded, revolutionizing both experimental and computational investigations. MXenes have emerged as key 2D nanomaterials due to their unique properties, making them focal points for computation-driven innovation (Fig. 2). The integration of advanced AI models has significantly accelerated the discovery, synthesis, functionalization, production scale-up, and translation into clinical application of MXenes¹⁴⁷. Further, to methodically analyze the integration of AI-based and computational simulations in MXene research, we conducted a meta-analysis using the Scopus database. Our search strategy employed (Fig. 7A) the keywords “mxene* AND (“machine learning” OR “deep learning” OR “predictive modeling” OR “computational simulation” OR “Density functional theory”) AND (“synthesis” OR “application” OR “sensors” OR “memristors” OR “actuators”)), applying Boolean logic to ensure comprehensive coverage of studies related to predictive modeling, computational and AI-based simulations in MXene synthesis and applications. The search was confined to research articles and reviews published in English between 2015 and 2025, aiming to identify emerging patterns, methodologies, and applications in MXene synthesis and functional deployment, especially in sensors, memristors, and actuators. This search yielded 1242 articles and 94 reviews, showing a surge in research interest over the last five years (Fig. 7B). The outcome of this

analysis revealed a dataset containing 1000 items grouped into 6 clusters, forming 89,963 links with a total link strength of 184,166, and was analyzed to guide the discussion in this section. Bibliometric assessment was carried out using VOS viewer, with “co-occurrence” as the analysis type and “author keywords” as the unit of analysis. Keyword occurrences were measured using the full counting method to ensure equal weighting for each keyword. The association strength method was applied to normalize relationships and enhance interpretability. The results were illustrated with density visualization (Fig. 7C) to highlight the prevailing research directions that lay the foundation for advancing future AI-driven biomedical innovations. Additionally, the bibliometric search was refined to a biological context by limiting the subject areas to “Biochemistry, genetics and Molecular Biology”, “Medicine”, and “Pharmacology, Toxicology and Pharmaceutics”. This resulted in the identification of 128 research articles and 11 reviews, enabling a targeted examination into the application of AI-driven and computational simulations within the bioengineering domain of MXenes.

AI-powered techniques, including high-throughput computational screening, autonomous discovery platforms, and machine learning-guided synthesis optimization, have notably improved the accuracy of property prediction, synthesis reproducibility, scalability of production, and optimization of MXene surface terminations for specialized applications. From a broader computational perspective, integration of techniques like density functional theory (DFT) combined with AI contributes to predicting and explaining the physicochemical properties and applicability of both experimentally synthesized and hypothetical MXenes¹⁴⁸. By unraveling the atomic and molecular intricacies of MXenes, these approaches deepen our fundamental understanding, driving the development of new nanomaterials with tailored properties, deciphering applicability, and guiding experimental synthesis processes. Leveraging AI's foundational role in analyzing existing data and simulating innovative approaches to conventional synthesis methods, machine learning (ML) and deep learning (DL) techniques are attaining prominence in advancing MXene research. By incorporating ML models trained on experimental and DFT-generated data into the synthesis of MXenes, these models facilitate the forecast of key MXene parameters and properties, including the definition of transition metal, “A” group elements, carbo-nitride forms, and the design of new MXene-derivatives optimized for specific uses^{149,150}. ML models, like random forest (RF), support vector regression (SVR), multiple linear regression (MLR) and artificial neural networks (ANN) are employed to predict MXene performance and stability by tuning synthesis parameters and stimulating MXene composition^{151–155}. Furthermore, ML identifies patterns within large input datasets to forecast the electronic, optical, thermal, and mechanical properties of MXenes, significantly accelerating material screening to prioritize promising candidates for experimental validation and reducing the experimental workload. Some of the databases developed to support this ML-driven exploration are the aNANt MXene database, which catalogs over 23,000 MXene structures with computed properties¹⁵⁶, the MEM-CES MXenes Database¹⁵⁷, offering over 730 structural files, and the MXene-db¹⁵⁸, which compiles structural and computational band gap data of over 4000 MXenes. While these resources have proven critical for energy storage applications, they are now progressively supporting biomedical application predictions, which are discussed in the upcoming section. Deep learning (DL) techniques, like convolutional neural networks (CNN), further enhance the analysis of complex material characterization data, enabling deeper insights into the MXene structure and property patterns from complex spectroscopy (FTIR, XPS), microscopy (SEM), and diffraction (XRD, EDX) datasets¹⁵⁹. This demonstrates that AI-driven computational modeling enables virtual screening across diverse MXene chemical compositions and structural specifications, facilitating the discovery of new MXenes with customized properties for targeted applications. Multi-target regression and classification models have identified optimized MXene compositions with gravimetric capabilities for energy-based applications¹⁶⁰, and AI-assisted automated workflows accelerate the experimental synthesis of MXene-based conductive aerogels, with tailored mechanical and electrical properties¹⁶¹. Notably, AI can also analyze extensive literature, patents,

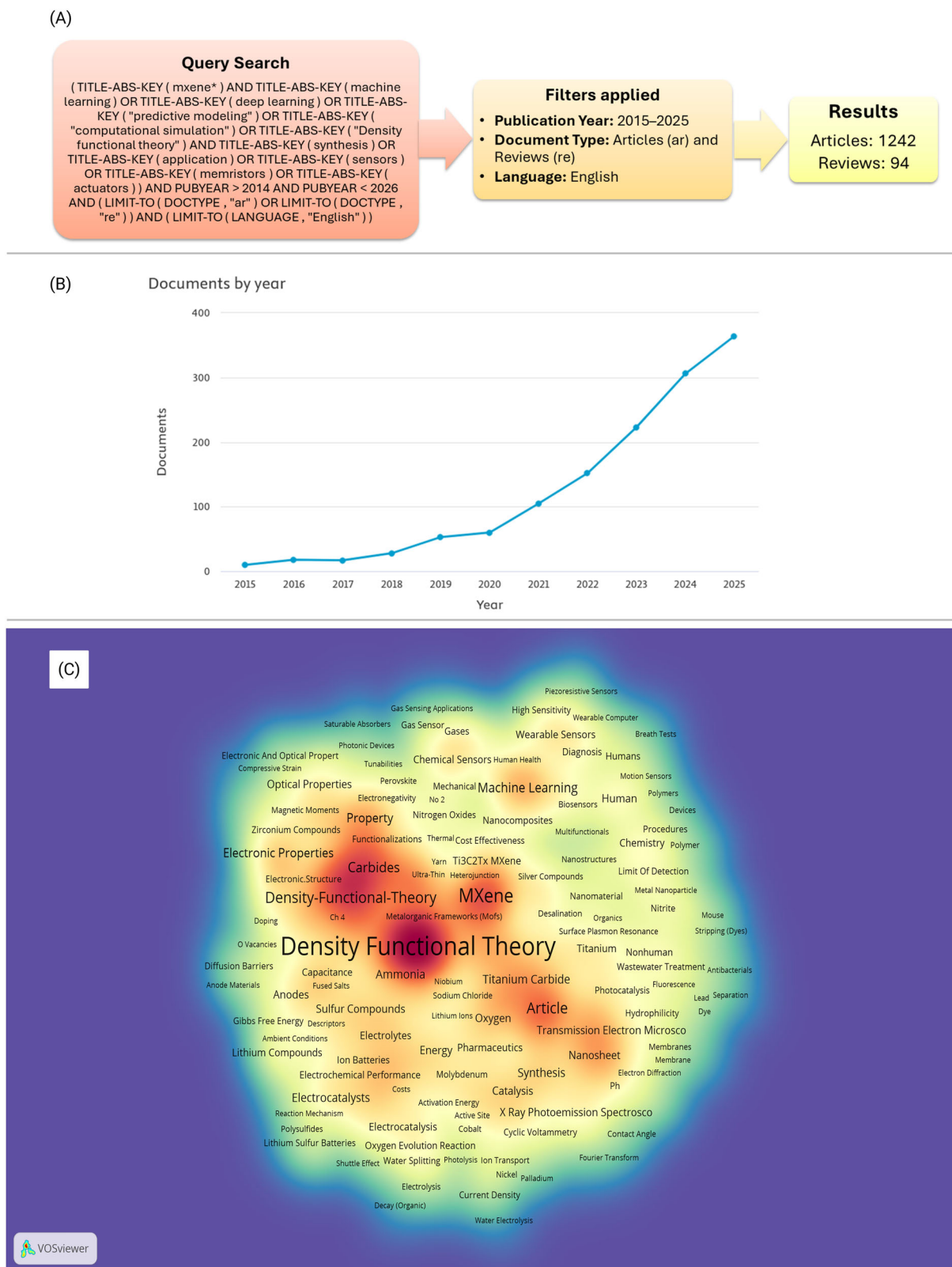


Fig. 7 | Overview of the meta-analysis conducted on the integration of predictive modeling, computational, and AI-driven simulations in MXene Research.

A Flowchart illustrating the literature search strategy, beginning with the initial query search and progressing through the applied filters. The query search was limited to articles and reviews published in English between 2015 and 2025, yielding 1242 articles and 94 reviews; **B** Distribution of publications by year over the last

decade; C Density visualization generated by VOS Viewer of keywords co-occurrence and thematic clusters in literature relating to MXene synthesis, computation, and applications. The analysis highlights emerging patterns, methodologies, and uses of predictive modeling, computational techniques, and AI-based simulations in MXene synthesis and deployment, with particular emphasis on sensors, memristors, and actuators.

and process experimental databases to generate a comprehensive knowledge repository that shapes new synthesis strategies and guides future research directions.

Specifically, our meta-analysis and bibliometric evaluation (Fig. 7) reveal a rapid expansion of AI-driven MXene-based biosensors, memristors, and actuators in biological applications. AI and deep learning techniques have been employed to develop highly sensitive MXene-based devices, including sound detectors¹⁶² and artificial eardrums¹⁶³ capable of precise voice recognition. Subjecting the deep learning network model to extensive training and testing using data collected from the MXene-based acoustic detector resulted in accurate detection of vowel sounds in human speech. The approach demonstrates the powerful synergy between MXene-based acoustic sensors and AI algorithms in enabling precise voice recognition systems. Stretchable MXene composite hydrogels have been engineered as wearable sensors for human motion detection, offering rapid responsiveness and durability for healthcare and defense mechanisms¹⁶⁴. This conductive composite could endure tensile strains exceeding 1800%, while serving as an adaptable wearable sensor for real-time detection of diverse human movements. Furthermore, an ultra-stretchable MXene-based organohydrogel (M-OH) with exceptional conductivity was prepared for health monitoring and enhanced object recognition through machine learning integration, achieving high sensing accuracy¹⁶⁵. This study demonstrated that the highly conductive nature of the M-OH enabled superior performance in health monitoring and object sensing applications. This advancement unlocks numerous opportunities for innovating in wearable technology using AI-based interfaces in personal healthcare. AI-driven MXene-based biosensors have shown promising advancements in disease detection and health monitoring. Innovations like black phosphorus-MXene composites have improved the deformability and pressure sensitivity of flexible biosensors¹⁶⁶. Multifunctional MXene-based patches integrated with AI facilitate real-time sweat analysis for biomarkers such as glucose and lactate, providing continuous health data through wireless transmission. MXene-based actuators and biosensors further enhance bioengineering applications by enabling artificial reflexes and sensitive biomarker detection, driving innovations in soft robotics, prosthetics, and personalized healthcare¹⁶⁶. Neuromorphic computing has the potential to revolutionize AI-enabled computing methods to mimic brain-like cognitive functions. The integration of Optical Neural Networks (ONNs) with MXenes represents a significant advancement in this field. Combining ONNs' inherent parallelism and low energy consumption with MXenes' tunable optical properties and electrical conductivity positions them as efficient ONN candidates for energy-efficient AI-based computing technologies¹⁶⁷. MXene-perovskite hybrid sensors demonstrate high responsivity and broad wavelength detection, promising advances in biomedical imaging¹⁶⁸. MXene-based memristors enable advanced synaptic functions, with their electrochemical behavior facilitating artificial memory enhancement, Pavlovian learning, and logic operations, alongside high sensitivity and dendritic integration¹⁶⁹. This study introduces an innovative approach for developing ultrasensitive artificial neural systems, enabling precise perception monitoring and control via advanced memristor devices. Meanwhile, MXene-based actuators hold significant promise in diverse smart-assistive technologies ranging from interactive human-machine interfaces and artificial prosthetics to assistive robotics and smart city applications¹⁷⁰. These reports demonstrate that AI-enabled MXenes technologies are transforming bioengineering and wearable technologies by enabling precise health monitoring, biosensing, and energy-efficient computing to support innovations in next-generation human-machine interfaces and biomedical devices.

Bibliometric analysis has also proven valuable in predicting and analyzing the biomedical applications of MXenes. It serves as a powerful research tool, especially with 2D nanomaterials in biomedical fields, to statistically evaluate and visually map the current landscape, key contributions, and emerging trends within the scientific field^{171,172}. By conducting quantitative and qualitative assessments of metrics such as countries, institutions, authors, and keywords from published literature, bibliometric

analysis offers a clear visualization of past research hotspots, leading contributors, and potential future directions. For instance, Saravanan et al. performed a comprehensive scientometric review on MXene research from 2012 to 2020, and their findings identified sensing and photocatalysis as major areas of focus, with China and the USA as leading contributors, and journals like ACS Applied Materials & Interfaces and Journal of Materials Chemistry A emerging as prominent publication platforms¹⁷³. Similarly, Guo et al. conducted a bibliometric study specifically examining MXenes' biomedical applications, revealing a concentration on MXenes' antibacterial uses by leveraging their photothermal properties, while predicting a future shift toward wearable electronics for medical purposes¹⁷⁴.

In line with these studies, we have incorporated a focused bibliometric analysis using VOSviewer in the present review (Figs. 3, 4, and 7). Titles, abstracts, and keywords were extracted from the meta-analysis to generate network visualizations, mapping key research concepts and application areas. These visualizations helped in the identification of core research themes, conceptual clusters, and emerging trends in the evolving field of MXene-based biomaterials. The resulting density and network maps provide an intuitive overview of the interconnectedness and relative importance of research topics within this curated literature set, offering valuable insights into the structure and evolution of the biomaterials field. Thus, the use of AI models in complement with bibliometric analysis provides valuable insights into advancing MXene discovery, synthesis, functionalization, research trends, and emerging medical applications. Together, they guide future investigations for the strategic development of various MXene-based technologies.

Tailored synthesis and properties of different transition metal-based MXenes for bioengineering applications

Since the discovery of the first MXenes derived from titanium-based MAX phases, various types of MXenes have been developed by incorporating different transition metals into their structure. As the “M” in “MXene” refers to a family of nanomaterials with the general formula $M_{n-1}X_nT_x$, where “M” refers to an early transition metal from groups 3 to 7 from the periodic table. In this section, we will dive deeper into the different types of transition metal-based MXenes, focusing on their synthesis method, unique properties, characterization methods used to identify these properties, and their potential applications in the field. Table 1 provides a summary of the various MXenes discussed in this section with focusing on the characterization techniques employed for each in relation to its specific synthesis and application.

Titanium-based MXenes

Owing to the excellent electrical conductivity, high surface area, and biocompatibility, titanium-based MXenes were the first of their kind to enter biomedical applications. The highly tailorable surface chemistry of titanium-based MXenes has proven viable for further surface modifications to be employed in antibacterial, drug delivery, bioimaging, biosensing, cancer theranostics, and tissue engineering applications. LiF/HCl etched $Ti_3C_2T_x$ powders were subjected to ultrasonication to obtain single-to few-layered nanosheets for exhibiting dose-dependent antibacterial activity for use in water treatment and biomedicine³². $Ti_3C_2T_x$ nanosheets measuring 200 nm in size and obtained by ultrasonication of precursor nanosheets, were surface modified with polymethacrylic acid with enhanced drug loading efficiency of doxorubicin, for synergic photothermal therapy and anticancer drug delivery³⁸. The use of mechanical delamination for MXene preparation, combined with surface functionalization, enables MXenes to serve dual applications in both drug delivery and cancer therapy. Dimethylformamide (DMF)-intercalated $Ti_3C_2T_x$ MXene, coupled with solvothermal treatment, yielded graphene quantum dots of high quality. These MXene-derived quantum dots exhibited robust photoluminescence and outstanding water dispersibility, making them suitable for cellular imaging applications that leverage their optical properties³⁹. Furthermore, by conjugating single-layered Ti_3C_2 MXenes with a fluorescence resonance energy transfer aptasensor, MXenes found applications in both biosensing and

Table 1 | Overview of transition metal-based MXene nanosystems: synthesis, characterization, and biological applications

MXene nanosystem	Synthesis method	Characterization techniques	Biological applications	Refs.
$\text{Ti}_3\text{C}_2\text{T}_x$ nanosheets	LiF/HCl etching followed by ultrasonication under argon gas	SEM, TEM, EDS, XRD	Antibacterial activity	32
$\text{Ti}_3\text{C}_2\text{T}_x$ -CoNWs (cobalt nanowires)	LiF/HCl etched MXenes and reduction method based CoNWs, mixed and sonicated in ice to obtain the nanosystem	SEM, TEM, XRD, AFM	Photothermal Property, Drug Loading, and Release	209
$\text{Ti}_3\text{C}_2\text{T}_x$ MXene Nanosheets	LiF/HCl etched ultrathin MXenes conjugated with doxorubicin	SEM, TEM, HRTEM, EDX, XRD	Drug delivery application in cancer treatment	38
Graphene quantum dots derived from Titanium carbide ($\text{Ti}_3\text{C}_2\text{T}_x$) MXene	HF etched MXenes subjected to solvothermal treatment in DMF	TEM, AFM, XRD, XPS, FTIR, Photoluminescence, and Raman spectroscopy	Biomaging Application	39
$\text{Ti}_3\text{C}_2\text{T}_x$ MXene Nanosheets	HF etched MXenes	SEM, XRD	Biosensing applications	175
$\text{Ti}_3\text{C}_2\text{T}_x$ MXene Nanosheets	Series of mechanical delamination techniques of probe and bath sonication	SEM, EDS, TEM, FFT, SAED, XRD, XPS	Immunomodulatory application in preventing graft rejection.	37
Ta_4C_3 MXene Quantum Dots	NaF/HCl etched MXene nanosheets subjected to mechanical delamination and hydrothermal process	TEM, HRTEM, SAED, FFT, SEM, EDS, XRD, XPS, FTIR	Immunomodulatory activities	36
Ta_4C_3 MXenes nanocomposite	HF etched MXene nanosheets with iron oxide nanoparticles grown in situ on the MXene surface, encapsulated with soybean phospholipid	TEM, HRTEM, XRD, EDS, XPS, FTIR	Photothermal therapy for breast cancer	184
ZrC MXene Nanosheets	Liquid-phase exfoliation method	TEM, HRTEM, SAED, AFM, XPS, XRD, FTIR	Cancer treatment	186
V_2CT_x MXene	HF etched MXenes delaminated with TBAOH, conjugated with PB and AuNPs	SEM, EDX, XPS, Raman Spectroscopy	Detection of IL-6 released from breast cancer cells	194
Nb_2CT_x Nanosheets	Chemical Exfoliation	SEM, TEM, EDS, AFM, XRD, Raman Spectroscopy	Anti-inflammatory actions, Prevention of Osteolysis, Biocompatibility, photothermal ability	195, 197
Cr_2CT_x MXene	Etching aluminum from the Cr_2AlC MAX phase using a hydrothermal method	SEM, XRD	Anti-microbial and antioxidant properties	200
WO_3/MXene nanocomposite	Prepared by combining WO_3 with MXene using a sonication technique	XRD, FESEM, EDX, FTIR, Brunauer-Emmett-Teller (BET). BET results predicted increased surface area and enhanced pore properties in the composite compared to individual components.	Antibacterial activity.	207
$\text{Mo}_2\text{Ti}_2\text{C}_3\text{T}_x$ Double transition metal MXenes	HCl/LiF method of etching followed by exfoliation using TBAOH	TEM, XRD, XPS, photothermal conversion efficiency	Antibacterial, wound healing, and anti-inflammatory properties	203
$\text{TiV/NbMoC}_3\text{T}_x$ high-entropy MXene (HE-M) with polydopamine	HE-M nanosheets prepared from commercially procured $\text{TiV/NbMoC}_3\text{T}_x$ by aqueous-phase exfoliation, followed by differential centrifugation	DLS, Zeta potential, TEM, EDS, SEM, AFM, photothermal conversion efficiency	Antibacterial and anticancer properties via photothermal and sonodynamic modalities to eliminate bacteria and tumors, respectively	204

clinical diagnosis applications¹⁷⁵. This MXene-based aptasensor was able to detect thrombin, a bioanalyte in human serum, with high specificity and sensitivity. The electrical property of MXenes makes them ideal for tissue engineering applications, where these MXenes, as conductive nanomaterials, mimic the natural electrical cues found at the damaged tissue sites. By providing exogenous electrical stimulation, MXenes-based scaffolds and hydrogels regulate cell behavior, promoting repair and regeneration, which enhances the overall tissue healing outcomes in skin⁴², nerve¹⁷⁶, cardiac¹⁷⁷, and bone tissue engineering. Particularly in bone regeneration, MXenes surface functionalized with hydroxyapatite create an ideal environment for cell attachment, proliferation, and bone cell differentiation¹⁷⁸. Additionally, free-standing single- to few-layered MXene nanosheets and quantum dots obtained through mechanical delamination and hydrothermal processes have shown promise in immunomodulatory applications³³, such as preventing graft rejection³⁷ as well as antiviral applications by mitigating viral infections⁴¹, respectively.

Titanium-based MXenes have served as the foundational models for most AI-based simulations, guiding the applications of both these first-discovered MXenes and newly developed ones across various fields. Innovations like MXene-integrated artificial eardrums¹⁶³ and wearable hydrogel sensors¹⁷⁹ have demonstrated that ML-driven acoustic signal sensing and human motion detection are achievable. Further advancements like the development of ultra-stretchable, highly conductive MXene-organohydrogel systems with ML-assisted sensitive object recognition and health monitoring, highlight the extensive potential of ML-based MXene algorithms in wearable technologies for personalized healthcare applications¹⁸⁰. MXenes have also been instrumental in neuromorphic systems, enhancing AI-based robotics, healthcare diagnostics, and cybersecurity through improved pattern recognition and adaptive responses prediction¹⁸¹. Additionally, the integration of optoelectronic functionalities into MXene-based artificial neuron, combined with ML-algorithms has led to faster, multitasking sensory systems, paving the way for the next generation of hardware-based AI advancements¹⁸². Collectively, these studies highlight how different synthesis methods contribute to various biomedical applications of titanium-based MXenes and how ML-assisted approaches are enhancing healthcare technologies.

Tantalum-based MXenes

Tantalum-based MXenes, which closely resemble titanium, offer enhanced stability, resistance to oxidation, mechanical strength, biocompatibility, and bioactivity^{35,183}. HCl/NaF etched tantalum carbide (Ta_4C_3) MXenes, in the form of quantum dots produced through a series of mechanical delamination methods, have demonstrated excellent immunomodulatory properties. These MXenes are encapsulated by the endothelial cells and help reduce the activation and proliferation of cytotoxic T-cells, which are responsible for graft rejection. This size-based tailoring of MXenes, along with their abundant negatively charged surface terminations, increases their biological interaction and facilitates their internalization³⁶. HF etched Ta_4C_3 MXenes surface functionalized with iron nanoparticles and soybean phospholipid have been developed to exhibit enhanced bioimaging and photothermal performance¹⁸⁴. The incorporation of iron nanoparticles imparted superparamagnetic properties, enabling their utilization as contrast agents for magnetic resonance imaging (MRI). The presence of Ta further contributed to computed tomography (CT) imaging, while functionalization with soybean phospholipid ensured superior biocompatibility. This MXene-based nanosystem showed complete tumor eradication without recurrence, highlighting the potential of tantalum-based MXenes in cancer diagnosis and treatment. Similarly, $\text{Ta}_4\text{C}_3\text{T}_x$ MXenes incorporated with manganese oxide nanoparticles exhibited improved photothermal properties and potential for photoacoustic imaging. Thus, tantalum-MXenes, with their enhanced stability and versatile surface functionalization opportunities, show significant potential in a range of biomedical applications, positioning them as a promising platform for advancing diagnostic technologies and improving therapeutic outcomes.

Zirconium-based MXenes

Zirconium-based MXenes, first reported in 2013, have been extensively studied and explored applications in corrosion resistance, energy storage, optoelectronics, environmental remediation, photonics, and recently, biomedicine¹⁸⁵. Zirconium carbide nanosheets, synthesized from bulk carbide through a series of probe and bath sonication processes, exhibit excellent photothermal properties and ultrahigh drug loading properties¹⁸⁶. This zirconium nanosystem was found to effectively target malignant cells, reduce tumor inflammation, and promote angiogenesis. Although relatively few biological applications of zirconium-based MXenes have been reported to date, computational studies utilizing Density Functional Theory (DFT) with the Perdew–Burke–Ernzerhof (PBE) functional have been instrumental in predicting their structural and electronic properties. PBE, a widely adopted exchange–correlation functional within DFT, facilitates the accurate modeling of zirconium MXenes, providing insights into their potential for bioengineering applications^{187–189}. In addition, DFT-based studies have accurately described the electronic ground state of Zr_2C MXenes¹⁸⁹ and highlighted their potential use as substrates for single-atom catalysts with controllable charge states¹⁹⁰. These computational advancements highlight the potential for future AI-driven exploration of zirconium-MXenes for bioengineering applications. Similarly, although hafnium-based MXenes have not yet been synthesized, they have attracted significant theoretical interest due to PBE-based predictions showing enhanced stability, following the successful realization of titanium- and zirconium-based MXenes¹⁹¹. These computational studies have further revealed the expected electronic properties of hafnium-based MXenes, demonstrating both metallic and semiconducting behaviors. Collectively, these findings underscore the growing potential of AI in predicting new MXene systems with tailored properties for specific applications.

Vanadium-based MXenes

Vanadium carbide (V_2CT_x) MXenes, as one of the non-titanium-based MXenes, have attracted significant attention in energy-based applications for their potential as electrode materials. Their large interlayer spacing, high conductivity, and tunable surface make them ideal for the synthesis of various metal-ion batteries, either as composites or heterostructures¹⁹². In addition to these energy storage applications, V_2CT_x MXenes also exhibit excellent biocompatibility and photothermal properties. Nanosheets of V_2CT_x MXenes prepared through algae extraction methods have shown strong near-infrared (NIR) absorption, positioning them as efficient photothermal agents for tumor ablation¹⁹³. This algae-based extraction in aqueous solution effectively intercalates and delaminates the vanadium MAX phase, to yield the MXenes nanosheets, showing this method can be used as a low-cost, high-yield, and environmentally friendly approach for MXene synthesis for biomedical applications. The tunable surface chemistry of V_2CT_x MXene also opens avenues for conjugation of various functionalities. For instance, a Prussian blue (PB) conjugated V_2CT_x MXene loaded with gold nanoparticles (Au-NPs) has been developed as an immunosensor for detecting interleukin-6¹⁹⁴. The conjugation with PB enhances MXenes' electrochemical properties, while the incorporation of Au-NPs aids in the conjugation of antibodies, resulting in enhanced analytical performance. The V_2CT_x MXenes were synthesized using the conventional HF etching route, followed by TBAOH delamination under nitrogen protection and sonication, to yield monolayers of V_2CT_x MXene. These studies show that the tunable properties of V_2CT_x MXenes present their significant promise in offering versatile solutions in both energy storage and cutting-edge biomedical technologies.

Niobium-based MXenes

Since 2015, niobium-based (Nb_2CT_x) MXenes have been explored in a variety of applications, including energy storage devices like supercapacitors, electromagnetic shielding, electrical conductors, antimicrobial coatings, in 3D printing, sensing, and biomedical applications. Their high conductivity, electrochemical properties, mechanical properties, biocompatibility, hydrophilicity, and tunable surface make them highly versatile¹⁹⁵. In biomedical applications, Nb_2CT_x has demonstrated unique

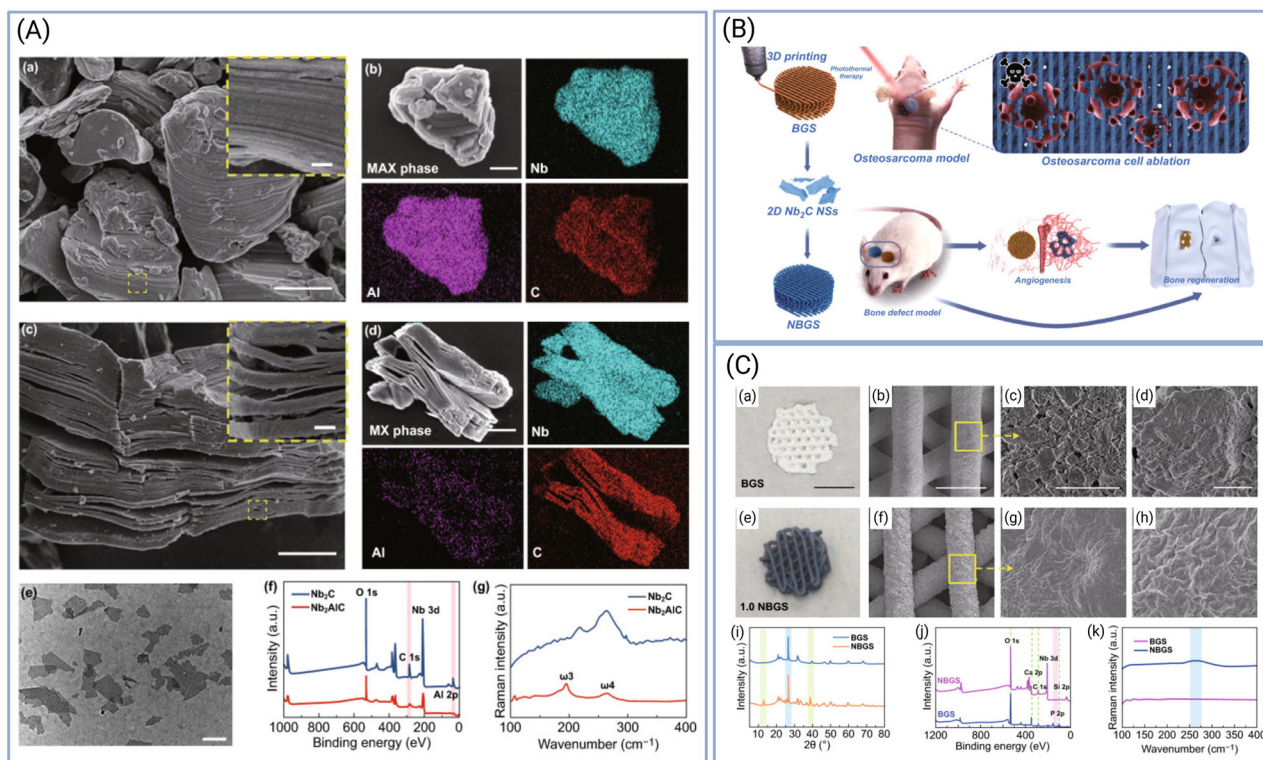


Fig. 8 | Synthesis and material characterization of ultrathin 2D Nb₂C MXene nanosystem. **A** (a, b) SEM image and EDS mapping of Nb₂AlC ceramics showing the presence of Nb, Al, and C. (c, d) SEM images and EDS mapping of multi-layered Nb₂C MXene (Scale bar for a–c = 1 μ m, and the bar of the inset = 100 nm). (e) TEM image of single/ few-layered Nb₂C MXene Nanosheets (Scale bar = 200 nm). (f) XPS spectra and (g) Raman spectra of Nb₂AlC MAX and Nb₂C Nanosheets. **B** Schematic

overview of synthesis and application of 3D NBGS. **C** (a–h) Digital photographs of bone-mimetic scaffold (BGS) with and without the Nb₂C Nanosheets (NBGS) (scale bar = 3 mm). (i) XRD patterns, (j) XPS spectra, (k) Raman survey of BGS and NBGS. (reproduced with permission from ref. 196 © The Author(s) 2020, corrected publication 2022).

characteristics like photonic response, antioxidant, and photothermal capabilities. Owing to this, Nb₂CT_x nanosheets incorporated into 3D printed porous bioactive glass scaffolds, find their application in treating osteosarcoma through photothermal ablation and promoting bone regeneration¹⁹⁶. Figure 8 shows the characterization of Nb₂CT_x-based nanosystem using SEM, TEM, XPS, XRD, and Raman spectra. Additionally, chemically exfoliated Nb₂CT_x nanosheets possess antioxidant activity by scavenging reactive oxygen species (ROS) and alleviating oxidative stress in osteolysis conditions¹⁹⁷. These MXenes were found to inhibit osteoclastogenesis and modulate macrophage activity after joint replacement surgery. Functionalization with polyvinylpyrrolidone enhances the ionizing ability and radioprotective activity of the MXenes by scavenging free radicals¹⁹⁸. Together, these findings underline the radioprotective, biocompatible, and bioactive nature of Nb₂CT_x, demonstrating their potential through both theoretical calculation and systematic in vitro and in vivo assessments.

Chromium-based MXenes

The interest in chromium-based (Cr₂C) MXenes stems from their potential to be modeled from theoretically predicted models based on pristine titanium-based MXenes. Using DFT calculations, Cr₂C MXenes with single-atom catalysts have been modeled¹⁹⁰. In this study, a systematic screening of nine transition metal-based bare MXene surfaces with M₂C stoichiometry was carried out, with chromium included as one of the investigated MXene metals. The findings revealed that the properties of the resultant single-atom catalysts can be tailored by appropriately selecting the MXene-metal combination, allowing control over their charge (partially oxidized or reduced) and demonstrating the feasibility of using MXenes as platforms for stabilizing single-metal atoms. Elsewhere, a systematic computational study employing DFT accurately described the electronic properties of M₂C MXenes, including Cr₂C, by analyzing their electronic band

structures to determine the electronic ground states¹⁸⁹. These computational studies predict that Cr₂C MXenes exhibit ferromagnetic-to-antiferromagnetic transition with a tunable energy gap and promising potential for nanoscale spintronics applications¹⁹⁹. Owing to this, recent efforts focusing on experimentally synthesizing Cr₂C MXenes were produced using LiF/NaF and HCl etchants with varying etching time and temperatures to identify optimal conditions for well-defined accordion structures. Further delamination was carried out with or without TBAOH to determine which method increased the yield and minimized byproducts. In this study, using SEM, EDX, XPS, XRD, and Raman spectroscopy, the characteristics of the synthesized Cr₂C MXenes were evaluated. There are limited studies in the literature on the synthesis and application of Cr₂C_x MXenes, showing the need for research on chromium-based MXenes to help advance the field. In such a venture, a very recent study was conducted to explore the antimicrobial properties of chromium MAX phases and MXenes²⁰⁰. Here, the MAX phase was synthesized using a pressureless sintering process, while the MXenes were produced by a hydrothermal method and thoroughly characterized. Upon this, the biological properties of chromium MAX and MXene were evaluated for the first time against two fungal and six bacterial strains, assessing the antimicrobial properties such as antioxidant activity, anti-biofilm activity, antidiabetic effects by measuring α -amylase inhibition, bacterial cell viability inhibition, and DNA scission. These findings highlight the multifunctional potential of chromium-based MXenes for a range of medical applications.

Molybdenum-based MXenes

Molybdenum-based (Mo₂C) MXenes are the next well-explored renewable energy-based MXenes after niobium, despite the low crystal abundance that these Mo₂C MXenes exhibit, flexibility to be synthesized from the MAX phases molybdenum aluminum carbide (Mo₂AlC) and gallium carbide

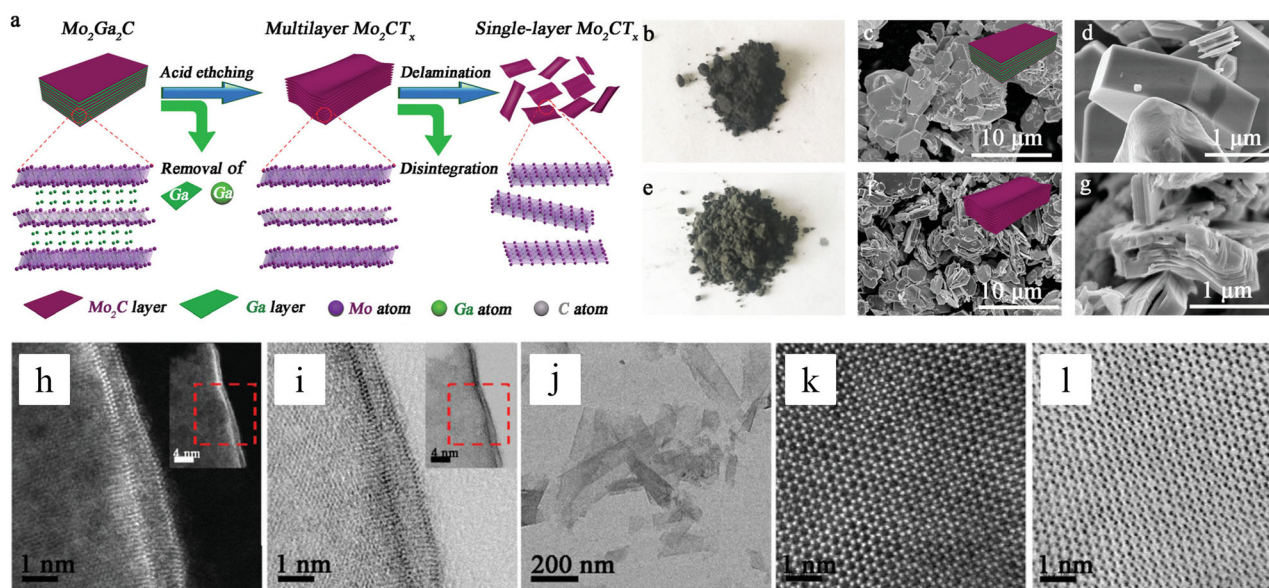


Fig. 9 | Synthesis and Characterization of Mo₂C MXenes. **a** Schematic overview of the synthesis and delamination process of Mo₂C MXenes. **b** Digital photo and **c, d** FESEM images of bulk Mo₂Ga₂C exhibiting a typical compact layered structure before HF etching. **e** Digital photo and **f, g** FESEM images of HF-etched multilayer

Mo₂C MXenes. **h, i** HAADF and ABF images of Mo₂C MXene after HCl/LiF etching. **j** Bright-field TEM image, **k** HAADF image, and **l** ABF image of HCl/LiF etched ultrathin Mo₂C MXene. (reproduced with permission from ref. 206. © 2019 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim.

(Mo₂Ga₂C) through top-down and bottom-up synthesis methods^{45,201}. Mo₂CT_x MXenes possess unique biocompatibility, photothermal (PT), photodynamic (PD), photoacoustic (PA), CT, and wound healing properties, showing their promise in cancer treatment²⁰². Mo₂Ti₂C₃T_x MXenes synthesized from Mo₂Ti₂AlC₃ MAX by the LiF/HCl etching and TBAOH delamination exhibited enhanced photostability and photothermal conversion efficiency. These double transition metal-based MXenes feature inner titanium layers sandwiched between outer molybdenum layers, exhibiting enhanced electrochemical properties. Alongside their photothermal properties, these MXenes exhibit strong antibacterial activity, showing their therapeutic potential in anti-inflammatory, antimicrobial, and wound healing applications, including effective treatment of bacteria-induced keratitis²⁰³. In addition, mixed transition metal-based high entropy MXene nanosystems, such as TiVNbMoC₃T_x, have been reported to exhibit high biocompatibility and strong photothermal properties²⁰⁴. These capabilities enable effective cancer treatment through photothermal-induced apoptosis, leading to tumor eradication upon laser irradiation. Mo₂C in the form of nanospheres was modeled through density functional theory to predict its electronic structure and photoabsorption behavior and synthesized to be tested with liver cancer cells to assess its photothermal ability and serve as a photosensitizer to generate ROS under light excitation for tumor treatment, thus serving as a cancer theranostic agent²⁰⁵. Well-characterized HF/TBAOH and LiF/HCl-etched Mo₂C MXenes (Fig. 9) surface functionalized with polyvinyl alcohol and employed as ultrathin nanoflakes served as an excellent platform for tumor ablation with efficient long-term storage using a vacuum package or inert gas shielding to elicit desirable physicochemical, biological, and therapeutic effects²⁰⁶.

Tungsten-based MXenes

Synthesis of tungsten (W)-based MXenes is challenging due to the predicted instability of their hypothetical MAX precursors, despite their promising low overpotentials in the hydrogen evolution reaction (HER). As a result, WO₃, as a metal oxide, has been doped with titanium MXenes to form a nanocomposite, leveraging their exceptional photostability and photocatalytic behavior. HF-etched Ti₃C₂ MXenes were sonicated with hydrothermally synthesized WO₃ nanorods to form the WO₃/MXenes nanocomposite and evaluated for their antibacterial and photocatalytic potential²⁰⁷. More recently, a theory-driven production of W-based MXene,

W₂TiC₂T_x, was developed from a non-MAX nano-laminated ternary carbide (W, Ti)₄C_{4-y} predecessor by selectively etching one of the covalently bonded W layers²⁰⁸. The study highlights the importance of W and Ti ordering, metal-layer vacancy defects, and oxygen-free carbon layers for effective selective etching of the precursor. Computational investigations and experimental evaluations were performed, and the W₂TiC₂T_x MXene exhibited high electrocatalytic HER performance (~144 mV overpotential at 10 mA/cm²). This study reports the synthesis of a W-based MXene from a covalently bonded non-MAX precursor, offering new synthetic approaches for 2D material synthesis. This sets the foundation for the future development of tungsten-based MXenes for direct biomedical applications.

Outlook

MXenes have revolutionized the field of biomedical engineering since their discovery, with exceptional potential in targeted drug delivery, photothermal therapy, imaging/diagnostic systems, antimicrobial applications, and tissue regeneration. As discussed in this review, their synthesis and tunable surface chemistry position them as transformative next-generation nanomaterials, but progress hinges on scalable, eco-friendly production of high-yield and quality MXenes. MXenes' clinical translation is dose-dependent and cell-type-specific toxicity and current experimental studies are limited to short-term evaluations. To address these challenges, advanced functionalization combined with long-term therapeutic efficacy evaluations and biocompatibility assessments is essential. It is needed to address these clinical challenges. Future research should prioritize refining MXene synthesis using fluoride-free etching and innovative bottom-up methods to optimize drug delivery, stability, and targeting. Long-term stability is particularly critical to prevent oxidation and preserve functionality, while reproducibility concerns arise from variations in synthesis protocols and surface modifications, which can impact batch consistency and clinical reliability. Although MXenes generally exhibit favorable biocompatibility due to their inert nature and hydrophilicity, specific studies report cytotoxicity or oxidative stress under certain conditions, emphasizing the need for comprehensive in vivo and chronic safety evaluations.

The meta-analysis presented in this review significantly contributes by systematically examining the progression of MXene synthesis techniques, their physicochemical attributes, and biological applications. The complementary meta-analysis focused on the convergence of MXenes

with AI-driven methods and computational modeling further enables a critical assessment of the field, underscoring both the promising opportunities and current challenges in predictive material design and simulation. Together, these comprehensive analyses provide valuable insights into the evolution of MXene research and highlight future directions for enhancing their design, synthesis, and bioengineering applications through AI-assisted approaches. Given that some MXene-based applications are still in early simulation stages, emphasis on theoretical calculations and AI-based simulations with complementary experimental work is essential for guiding MXene development. Clarifying the relationships between synthesis, structure, and resulting properties is central for tailoring MXenes for precise applications, particularly as exploration and validation of predicted functionalities in this fast-growing field, thereby elucidating the full potential of these materials. The integration of AI into MXene synthesis and application offers promising solutions to accelerate data-driven discovery of new MXenes, optimization of synthesis processes, increase yield, and enhance overall performance. AI-driven predictive modeling can forecast MXene properties based on elemental composition and processing conditions, directing researchers toward the most promising candidates and reducing reliance on exhaustive experiments. However, AI applications face challenges such as biases arising from limited or non-representative datasets and data scarcity due to insufficient experimental validation, which can impact model applicability and reliability. Overcoming these issues will be essential to fully harness AI's potential in advancing MXene research and its applications in biomedicine and technology.

In conclusion, MXenes offer great promise to transform the biomedical field, especially in nanomedicine. However, fully unlocking their potential hinges on overcoming existing challenges related to long-term biocompatibility, toxicity, scalability, and controlled drug-release. By integrating MXenes with diverse functional materials to create hybrid systems, therapeutic efficacy can be enhanced. A multidisciplinary approach, integrating advanced synthesis techniques, comprehensive characterization, and predictive modeling, will be vital for advancing MXenes from bench to bedside, ultimately improving diagnostics, therapeutics, and regenerative medicine strategies.

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Author contributions

The study was conceptualized and designed by L.R.S. and S.D. L.R.S., M.G., G.S., E.V., and S.D. drafted the manuscript and designed Figs. 1–4, and 7. All authors have read and approved the final version of this comprehensive review manuscript.

Competing interests

The authors declare no competing interests.

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